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Sec : 19

CSE340

Ans. to the ques. no. 1

(a)

0000 0000 1111 0010 1001 1010 1001 0011
└──────────┘ └──┬──┘ └──┬──┘ └──┬──┘ └──┬──┘
Immediate rs1 funct3 rd opcode

Opcode = 0010011 which is I-type

This is denoted by LD, LW, LH, LB, ADDI, SLLI, SRLI, ORI

Here,

funct3 = SLLI

rs1 = X₅

rd = X₂₁

Immediate = 16

∴ The RISC-V code - SLLI X₂₁, X₅, 16

(b)

0x089739B3

0000	1000	1001	0111	0011	1001	101	0011
funct7		rs2	rs1	funct3	rd	opcode	

Opcode = 0110011 which is of R-Type

funct3 = OR

rd = X₁₉

rs1 = X₁₄

rs2 = X₉

∴ The RISC-V code is — OR X₁₉, X₁₄, X₉.

Ans. to the ques. no. 2

Loop:

addi X5, X11, 13

addi X6, X13, -4

blt X5, X6, Exit

bge X12, X11, Elif

bge X12, X13, Elif

add X11, X12, X6

beq X0, X6, loop

~~Elif:~~

Elif:

addi X7, X0, 5

bne X11, X7, Else

addi X13, X13, 10

beq X10, X0, loop

Else :

addi X14, X14, 10

beq X0, X0, loop

Exit

Ans. to the ques. no. 3

Load and Store falls under I-type and S-type format respectively.

The funct3 field specifies the type and size of the data being loaded from the memory. It tells the processor how many bytes to load and how to interpret them. Load belongs to I-type format and their types are -

lb = Load byte (8 bit)

lh = Load half word (16 bit)

lw = Load word (32 bit)

ld = Load double word (64 bit)

Ans. to the ques. no. 4

(a)

beg X_0, X_0 , loop is a SB type instruction.

$$rs1 = X_0 = 00000$$

$$rs2 = X_0 = 00000$$

$$funct3 = 011$$

$$\text{Opcode} = (103)_{10} = 1100111$$

$$\text{Jumping offset} = 7028 - 7056$$

$$= -28$$

$$\therefore (28)_{10} = 0000\ 0001\ 1100$$

$$\text{1bit extension} = \begin{array}{r} 00000\ 0001\ 1100 \\ 11111\ 1110\ 0011 \\ +1 \end{array}$$

$$\therefore (-28)_{10} = \begin{array}{r} 11111\ 1110\ 0100 \\ \hline \begin{array}{l} \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ 12\ 11\ [10:5]\ [4:1] \end{array} \quad \text{Discard} \end{array}$$

7028 slli $X_{10}, X_{22}, 3$

7032 Add X_{10}, X_{10}, X_{25}

7036 ~~Add~~ $X_9, 0(X_{10})$

7040 Addi $X_{17}, X_0, -7$

7044 Bne X_9, X_{26} Exit

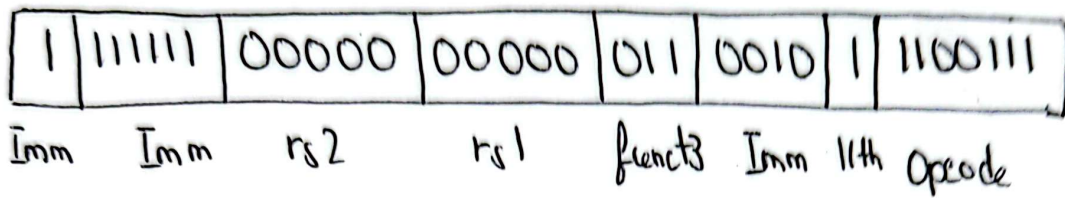
7048 Addi $X_{22}, X_{22}, 1$

7052 sd $X_{10}, 16(X_{22})$

7056 Beg X_0, X_0 , loop

Exit

7060 Empty line



(b)

Bne X9, X26, Exit

$$\begin{aligned} \text{Jumping offset} &= 7060 - 7044 \\ &= 16 \end{aligned}$$

$$rs1 = X9 = 01001$$

$$rs2 = X26 = 11010$$

$$(16)_{10} = 0000 \ 0001 \ 0000$$

$$\begin{array}{ccccccc} \text{1 bit extension} & = & 0 & 0000 & 0001 & 0000 & \\ & & \downarrow & \downarrow & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ & & 12 & 11 & [10:5] & [4:1] & \text{Discard} \end{array}$$

